

Customer Support Ticket Analytics System

Overview

The Customer Support Ticket Analytics System is a Python-based application designed to manage, analyze, and report customer support tickets. The system tracks tickets, categorizes them based on priority and issue type, and generates operational insights for support teams.

Features

Ticket Management

- Create and manage support tickets
- Track customer information
- Assign ticket priorities
- Categorize support requests
- Monitor ticket status

Analytics

- Priority distribution analysis
- Category-wise ticket analysis
- Support workload reporting
- Trend analysis support

Reporting

- CSV export functionality
- Structured ticket reporting
- Dashboard-ready outputs

Architecture

Core Components

Ticket Class

Represents an individual support ticket containing:

- Ticket ID
- Customer Name
- Priority Level
- Category
- Creation Timestamp
- Status

TicketAnalytics Class

Provides:

- Ticket aggregation
- Statistical analysis
- Report generation
- Data export

Sample Categories

- Billing Issues
- Technical Support
- Account Management
- Sales Inquiries

Sample Priorities

- LOW
- MEDIUM
- HIGH

Business Use Cases

- Customer Service Operations
- Help Desk Management
- IT Service Management
- Customer Experience Monitoring
- SLA Tracking

Execution

```
python ticket_analytics.py
```

Expected Output

```
Counter({'LOW': 167, 'MEDIUM': 167, 'HIGH': 166})  
Counter({'Billing': 125, 'Technical': 125, 'Account': 125, 'Sales': 125})
```

Future Enhancements

- Database Integration
- REST APIs
- SLA Monitoring

- Ticket Assignment Engine
- AI-based Ticket Classification
- Predictive Analytics
- Interactive Dashboards